

Mayank Kundalwal

Research Scientist | Federated Learning & Privacy-Preserving AI

Department of Computer Science & Engineering
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🌐 LinkedIn 🎓 Google Scholar (74+ citations)

Research Summary

PhD in Computer Science (IIT Jodhpur, thesis submitted) specializing in **federated learning, privacy-preserving machine learning, and medical AI**. Developed four novel FL frameworks addressing communication efficiency, label scarcity, adversarial robustness, and machine unlearning — published in **IEEE TAI**, **IEEE TETCI**, and **WACV 2026**. Experienced in building large-scale distributed ML systems under real-world heterogeneity constraints, with additional applied research on radar-based intelligent sensing for defense applications. Seeking a research scientist or postdoctoral position to advance privacy-aware and trustworthy distributed AI at scale.

Research Interests

Federated Learning, Federated Unlearning, Privacy-Preserving ML, Trustworthy & Robust AI, Medical Image Analysis, Machine Unlearning, Distributed AI Systems, Representation Learning

Education

- 2019–2025 **PhD, Computer Science & Engineering, Indian Institute of Technology Jodhpur**
Thesis Submitted Advisor: Dr. Deepak Mishra
Towards Robust Privacy-Preserving Federated Learning under Heterogeneous Environments for Medical Imaging Applications
- 2016–2018 **M.Tech, Computer Science & Engineering, National Institute of Technology Patna**
Thesis: Privacy-Preservation Framework for Cloud-Based Healthcare Systems CGPA: 8.24/10
- 2012–2016 **B.Tech, Computer Science & Engineering, Poornima University Jaipur**
CGPA: 7.38/10

Publications

Journal Articles

- 2025 **Mayank Kumar Kundalwal** and Deepak Mishra, FLAME: Federated Learning with Masked Autoencoders and Mean-Prototypes Embedding for Sparsely Labeled Medical Images, *IEEE Transactions on Emerging Topics in Computational Intelligence*
Addresses severe label scarcity in federated medical imaging using masked autoencoders and prototype-based embedding; enables high-accuracy FL with as few as 5% labeled samples per client.
- 2025 **Mayank Kumar Kundalwal** and Deepak Mishra, AR2FL: Anomaly-Resistant Robust Framework for Federated Learning, *IEEE Transactions on Artificial Intelligence*
Proposes a robust aggregation framework resilient to Byzantine and adversarial anomalies in heterogeneous FL settings; demonstrates consistent gains over FedAvg under IID and non-IID data distributions.
- 2019 **Mayank Kumar Kundalwal**, Kakali Chatterjee, and Ashish Singh, An Improved Privacy Preservation Technique in Health-Cloud, *ICT Express, Elsevier, Vol. 5, pp. 167–172*

Conference Proceedings

- 2026 **Mayank Kumar Kundalwal** and Deepak Mishra, Federated Model Synchronization for Diagnostic Redefinition through a Novel Selective Parameter Unlearning, *IEEE/CVF Winter Conference on Applications of Computer Vision (WACV 2026)*
Introduces selective parameter unlearning to enable compliant model updates without full retraining; directly addresses “right-to-be-forgotten” requirements in federated medical systems. **DEI Travel Grant Recipient.**

- 2024 **Mayank Kumar Kundalwal**, Anurag Saraswat, Ishan Mishra, and Deepak Mishra, Client Contribution Normalization for Enhanced Federated Learning, *IEEE 21st India Council International Conference (INDICON 2024)*
- 2022 **Mayank Kumar Kundalwal**, Anurag Saraswat, Ishan Mishra, and Deepak Mishra, BAFL: Federated Learning with Base Ablation for Cost Effective Communication, *26th International Conference on Pattern Recognition (ICPR 2022)*
- Reduces inter-client communication overhead in FL by selectively transmitting only base-layer parameters; achieves significant bandwidth reduction with minimal accuracy trade-off.
- 2018 **Mayank Kumar Kundalwal**, Ashish Singh, and Kakali Chatterjee, A Privacy Framework in Cloud Computing for Healthcare Data, *IEEE International Conference on Advances in Computing, Communication Control and Networking (ICACCCN 2018)*

Research Experience

- 2019–2025 **Doctoral Researcher — Federated Learning Systems, IIT Jodhpur, MAISys Lab**
- Designed and implemented four FL frameworks (BAFL, FLAME, AR2FL, SPU) in PyTorch, covering communication efficiency, semi-supervised learning, Byzantine robustness, and machine unlearning.
 - Developed theoretical convergence analyses and privacy guarantees for proposed frameworks.
 - Published 2 IEEE Transactions papers and 1 WACV paper; 2 manuscripts under review; 74+ Google Scholar citations.
- 2023–Present **Research Collaborator — Defense AI Sensing (Indian Army Sponsored), IIT Jodhpur**
- Developing real-time long-range moving target detection algorithms on pulse-Doppler radar data under adverse environmental conditions.
 - Responsible for radar signal processing pipeline, motion feature extraction, and robust ML-based detection model design.
 - Demonstrates applied experience in safety-critical, resource-constrained intelligent sensing systems.

Technical Skills

- Languages Python (primary), C/C++, Java, Bash
- ML PyTorch, TensorFlow, Keras, Flower (FL framework), scikit-learn
- Frameworks
- Distributed Systems Federated Learning architectures (FedAvg, FedProx, SCAFFOLD), Non-IID data partitioning (Dirichlet), Multi-client simulation
- Data & Tools NumPy, Pandas, Matplotlib, Weights & Biases, Git/GitHub, L^AT_EX, Linux/HPC environments

Honors & Recognition

- 2026 IEEE/CVF DEI Broadening Participation Travel Grant — WACV 2026
- 2025 Graduate Forum Poster Presentation — IndoML 2025
- 2023–2025 Defense Research Funding Support (Indian Army Sponsored Project)
- 2019–2023 Ministry of Education Doctoral Fellowship (formerly MHRD)
- 2016–2018 MHRD Fellowship, NIT Patna
- 2023–Present Cyber Volunteer Squad Member, Ministry of Home Affairs, Government of India
- Reviewer IEEE Transactions on Artificial Intelligence (TAI), IEEE TETCI, IEEE Internet of Things Journal, IEEE TDSC, WACV, ICPR, IJCAI, ECAI

Referees

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